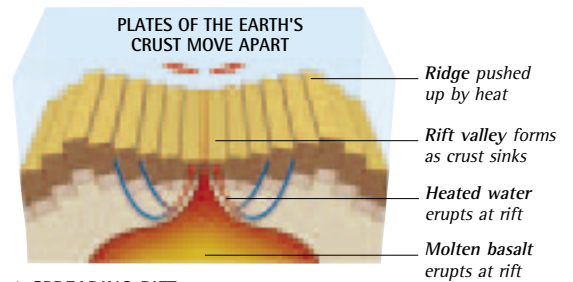


MIDOCEAN RIDGES

Ocean floor is formed where plates of oceanic crust move apart to create a spreading rift. This relieves pressure on the hot mantle rock below, and since pressure is all that keeps the hot rock solid, it turns into molten magma and squirts up through the rift as basalt lava. The lava cools in the cold ocean waters to form the solid rock of the ocean floor. Meanwhile, the magma chamber beneath the rift makes the rock expand, raising the crust on each side of the rift into two lines of submarine mountains that form a midocean ridge.

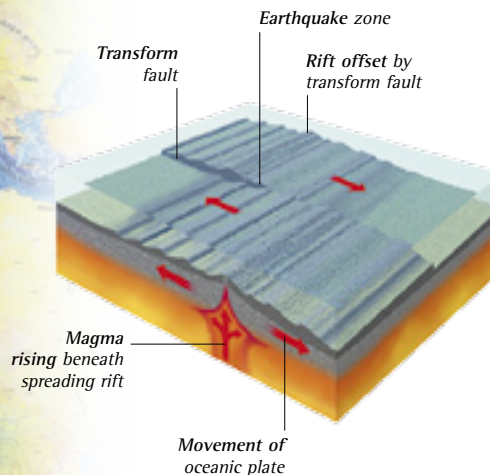


▲ SPREADING RIFT

Where the ocean floor is being pulled apart by convection currents in the hot mantle below, big blocks of oceanic crust sink to form a rift valley at the boundary. The floor of the valley is full of fissures, which erupt basalt lava and superheated water rich in dissolved chemicals. On each side of the rift, blocks of ocean crust are raised into ridges by the heat, but they gradually sink again as they slowly move away from the hot zone.

◀ WINDING RIDGES

The midocean ridges created by rifting and uplift extend for vast distances across the ocean floor. The Mid-Atlantic Ridge—seen here on the Heezen–Tharp map—extends for about 10,000 miles (16,000 km), all the way from the Southern Ocean to the Arctic Ocean. Its mountains are up to 13,000 ft (4,000 m) high, but their peaks lie 6,500 ft (2,000 m) beneath the surface of the ocean.



◀ TRANSFORM FAULTS

Midocean ridges are split into short sections divided by long transform faults that cut straight across them. These faults are very obvious features of the ocean floor map shown on the far left. They are caused by slightly different spreading rates along the rift. Where they link sections of the spreading rift, the ocean floor on either side of the fault is moving in opposite directions, and this causes submarine earthquakes.

◀ THE OPENING ATLANTIC

The winding route of the Mid-Atlantic Ridge echoes the shapes of South America and Africa. This is not a coincidence. Some 130 million years ago the two continents were joined together, but then a rift opened up between them, as seen here. The rift was flooded with water, and a new ocean floor formed in the widening gap to create the Atlantic Ocean.

